

Contract billing business logic

Standard Invoice, Progress Invoice, Change Orders, Retainage, Payment Schedule, and QuickBooks Online

Canonical handoff / September 2026

Business rules for product, design, engineering, QA, and AI review

System overview

One contract model supports tracked billing from Estimate acceptance onward.

Standard Invoice

Copies an Estimate into an independently editable Invoice.

Progress Invoice

Bills accepted Contract scope and tracks completion.

Change Order

Revises the current Contract while preserving history.

Retainage

Withholds part of an Invoice and releases it later.

Payment Schedule

Plans milestones without creating or reserving scope.

QuickBooks Online

Receives accounting documents, not Contract logic.

Canonical Contract model

An Accepted Estimate establishes the Contract. Accepted Change Orders revise the same Contract.



No ledger reset. No history migration. No Estimate-to-CO item mapping.

Shared financial measures

Measure	Business meaning	Includes / excludes
Contract subtotal	Aggregate accepted pre-discount source scope	Estimate plus Accepted CO changes and Cost Plus
Contract Value	Current accepted economic value	Contract subtotal less accepted Contract discount
Previously invoiced	Posted tracked billing before the current Invoice	Draft reservations excluded
Remaining subtotal	Gross Contract scope still available in creation	Posted allocations and active Draft reservations excluded
Contract Progress	Posted gross scope divided by current gross scope	Discount, tax, retainage, and payments excluded
Discount Pool	Contract discount still available to apply	Kept separate from Remaining subtotal

Remaining subtotal answers how much underlying scope remains. The Discount Pool answers how much discount remains.

Contract billing and document lifecycle

Billing status belongs to the current Contract.



A positive Accepted CO can move a Billed Contract back to Partially Billed.

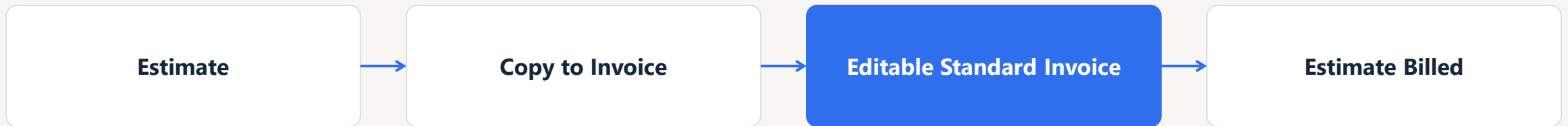
Change Order lifecycle

- Draft
- Pending
- Accepted
- Declined
- Canceled

Acceptance remains a fact even when a surface shows the Contract billing status.

Standard Invoice

Simple copy and conversion flow



Uses existing Standard Invoice rules

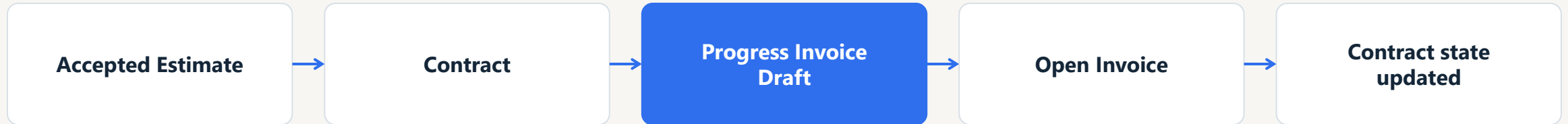
- Copied lines can change under existing rules
- Saving the Draft creates the conversion lock
- Deleting or canceling removes that lock when allowed

Separate from tracked Progress billing

- No Contract Progress allocation
- No source-line Progress ledger
- No milestone or Retainage Release behavior

One active Standard Invoice per Estimate; unavailable after tracked Progress history — enforce across creation, linking, API, import, and automation.

Progress Invoice happy flow



Draft

- Reserves selected source scope
- Reserves selected discount
- Does not increase posted progress

Open

- Consumes the reservation
- Updates Previously invoiced
- Updates Remaining subtotal and billing state

Progress always belongs to the Contract, including the first Invoice before any CO exists.

Progress Invoice creation

Supported paths use the same Contract availability rules.

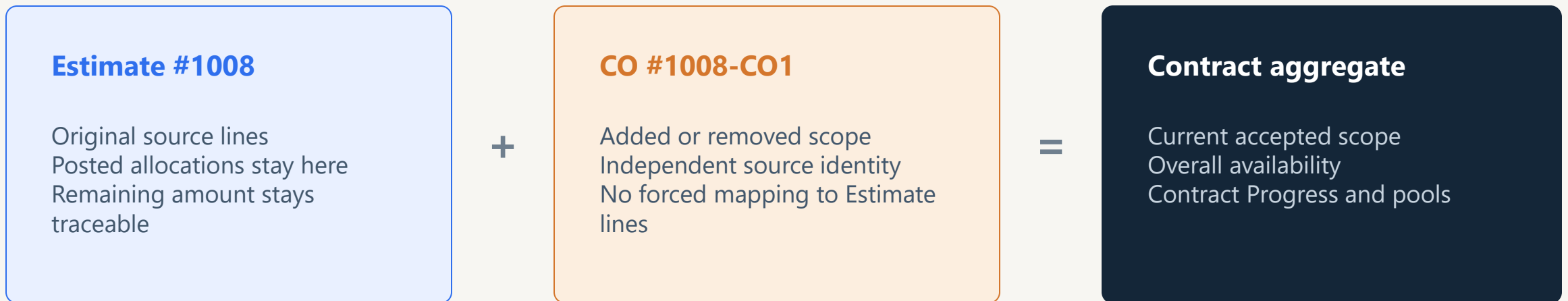
01	Full remaining Contract	Bills all eligible remaining source scope
02	Percent or amount	Allocates a portion within Contract limits
03	Select source lines	Bills selected eligible lines up to their remaining amount
04	Payment milestone	Uses planned context without bypassing availability

Remaining subtotal = gross Contract scope left after posted allocations and active Draft reservations

The Discount Pool remains separate.

Source-line lineage

The Contract aggregates scope without flattening its origin.



Contract revisions never rewrite historical invoice allocation.

Progress Invoice frontend requirements

Creation and editing

- Use current Contract scope in the creation prompt
- Show Remaining subtotal before discount
- Keep Discount Pool selection separate
- Group Estimate and CO lines when COs exist
- Keep Cost Plus computed and locked

State and controls

- Expose Draft reservation and one-Draft limit
- Lock prior invoices under existing chronology rules
- Show Contract Progress and billing status
- Prevent billing beyond Contract and line availability
- Preserve linked history and explicit navigation

Unsaved Invoice states do not show PDF preview or Customer web preview actions.

Progress Invoice customer presentation

Web link and PDF carry the same business information.

Items table

- Contract Amount
- Previously Billed
- This Invoice
- Balance to Finish
- Line completion when required

Presentation rules

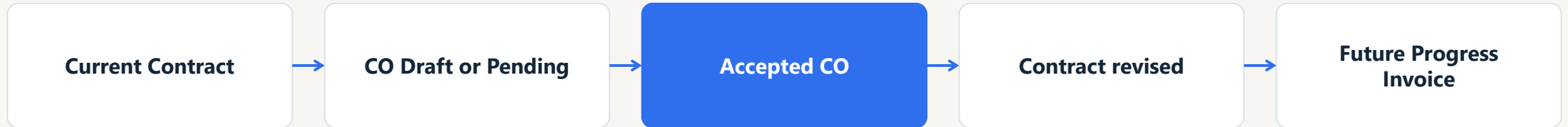
- Preserve item descriptions
- Emphasize billed rows
- Hide or soften unbilled rows
- Hide fractional quantities where required
- Exclude internal ledger mechanics

Cost Plus and totals

- One aggregate Cost plus fee line
- Description shows the percentage
- Place before subtotal
- Include fee in subtotal
- Do not duplicate it in totals

Tax, discount, Cost Plus, and retainage appear only when applicable. Zero-value placeholders stay absent.

Change Order happy flow



Acceptance updates

- Accepted source scope
- Contract totals and applicable pools
- Payment Schedule projections
- Contract billing status when scope reopens

Acceptance preserves

- Posted Invoice history
- Original source allocation
- Prior Contract Progress history
- Separate CO approval history

Change Order state and Contract effect

Draft	Proposed	Editable under existing locks
Pending	Proposed	Awaiting acceptance
Accepted	Included	Financially immutable
Declined	None	Historical
Canceled	None	Historical

CO
approval
state

Contract billing status can change after CO acceptance. CO lifecycle history remains intact.

Change Order document requirements

Internal document

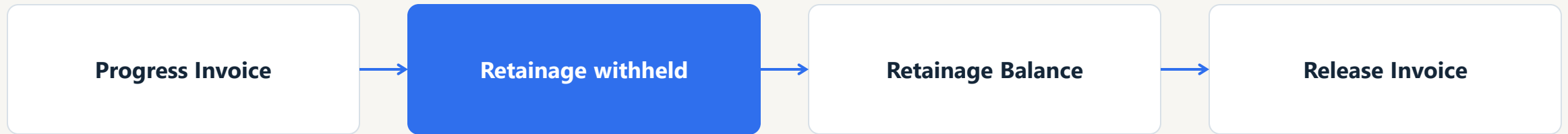
- Estimate-style editor with Change Order title
- Sequential number such as #1008-CO1
- Original Estimate and Contract reference
- Signed Contract change and shared summary
- Line-scoped conflicts; unrelated COs may coexist

Customer web link and PDF

- Reuse the Estimate customer document system
- Title the document Change Order
- Show Linked to Estimate #...
- Explain added, removed, or revised scope
- Use CO items, totals, notes, attachments, and terms

Internal pools and allocation mechanics never appear on the customer document.

Retainage happy flow



Withholding

- Reduces the Invoice amount due
- Leaves completed Contract scope unchanged
- Creates held Retainage Balance

Release

- Bills previously withheld money
- Supports partial or full release
- Creates no new Contract scope

Contract Progress and milestone completion use the full gross work billed.

Retainage frontend and customer requirements

Internal controls

- Show withholding only in enabled scenarios
- Display tracking only after retainage exists
- Track retained, released, held, and available amounts
- Block release above available balance
- Keep one active release Draft under existing rules

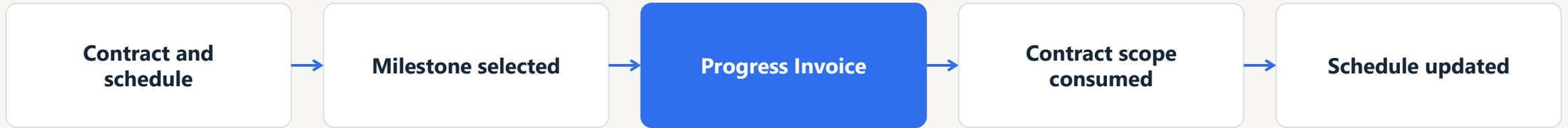
Customer presentation

- Original Invoice shows Retainage as a deduction
- Release Invoice identifies previously withheld funds
- Single line: Retainage release
- Qty 1 and the released amount
- No suggestion that release represents new work

A Retainage release calculates no second tax and does not increase Total Invoiced.

Payment Schedule model

A schedule plans billing milestones on the Contract. Actual Progress Invoices remain authoritative.



A milestone can guide amount and naming.

It cannot bypass source-line availability, Draft reservations, or overbilling protection.

Schedule state follows the linked Invoice.

Planned becomes Draft invoice, then Invoiced when the Invoice opens. Cancellation returns the milestone to Planned.

Schedule allocation can exceed 100%. The warning does not create scope or authorize billing.

Payment Schedule interactions

Standard Invoice	Separate conversion path	Successful save removes the applicable schedule path under existing rules
Progress Invoice	Milestone guides the draw	Actual allocation consumes Contract scope
Accepted CO	Contract Value changes	Future percentage milestones recalculate; historical actuals stay fixed
Retainage	Withholding stays separate	Release does not create milestone work

Historical milestone amounts never rebalance automatically after actual invoice variance.

Progress Invoice and Change Order edge cases

Active Draft exists

Block another tracked Draft; keep reserved scope unavailable.

Attempted overbilling

Prevent creation beyond Contract and source-line availability.

Final scope with unused discount

Warn about unused Discount Pool and resulting over-invoiced amount.

Posted Invoice edit

Only the latest eligible unpaid Open Invoice can change financially.

CO accepted with Draft

Cancel the Draft, release reservations, and require a fresh draw.

Positive CO after billing

Return the Contract to Partially Billed.

Deductive CO

Keep history fixed; show over-invoiced state when posted billing exceeds revised scope.

Accepted CO correction

Create another CO. Do not edit accepted financial scope.

Tax Credit remains separate from principal when a deductive CO removes previously taxed scope.

Retainage and Payment Schedule edge cases

Retainage

- No retainage means no tracking UI or totals row
- Partial release leaves the remainder held
- Over-release stays blocked
- Full Contract completion can coexist with held retainage
- Invoice cancellation reverses recognized withholding under existing rules

Payment Schedule

- More than 100% remains advisory over-allocation
- Draft-linked milestone is not fulfilled
- Invoiced milestone cannot create another Invoice
- Accepted CO recalculates future projections
- Canceled Invoice restores the live milestone to Planned

QuickBooks Online ownership boundary

Cinderblock owns Contract billing state

- Contract Progress
- Source-line allocations
- Discount and Cost Plus state
- Retainage Balance
- Payment Schedule and CO relationships

QBO receives accounting documents

- QBO Estimates
- Ordinary QBO Invoices
- Actual quantities and amounts
- Represented discounts and fees
- Tax facts compatible with QBO configuration

Cinderblock does not use QBO Progress Invoicing or QBO Estimate-to-Invoice progress relationships.

QBO document mapping

Cinderblock	QuickBooks Online	Business rule
Estimate	Standard QBO Estimate	No QBO Progress Invoicing
Change Order	Separate QBO Estimate	Preserve CO identity; do not modify original Estimate
Standard Invoice	Ordinary QBO Invoice	Use existing Invoice sync behavior
Progress / Contract Invoice	Ordinary QBO Invoice	No QBO Estimate linkage or Contract lineage
Retainage Release Invoice	Ordinary QBO Invoice	One positive Retainage release line

Status mapping: Partially Billed → QBO Converted; Billed → QBO Converted. Cinderblock preserves the local distinction.

QBO line representation

Progress quantity

Contract line
Qty 10 × \$100

25% billed
Qty 2.5 × \$100 = \$250

QBO receives the actual fractional quantity even if the customer view hides it.

Retainage withheld

Retainage	Qty 1	-\$1,000
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One negative line. No allocation across work lines.

Retainage release

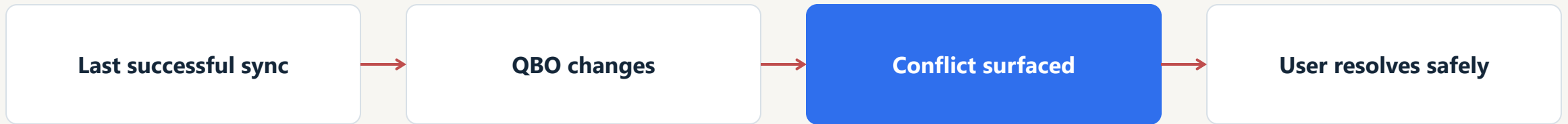
Retainage release	Qty 1	\$1,000
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One positive line. No original work lines are resent.

Discount and Cost Plus sync only when the Cinderblock document represents them.

QBO accounting safeguards

A diverged QBO transaction requires an explicit conflict state.



Protected situations

- QBO edit, deletion, or void
- Payment or credit recorded in QBO
- Cinderblock edit, cancellation, or invalidation
- Failed or uncertain synchronization

Required outcome

- Preserve Cinderblock Contract state
- Do not silently overwrite or recreate QBO
- Block duplicate Accounts Receivable risk
- Require intentional accounting resolution

QBO payment activity can change settlement state. It never changes Contract Progress or Contract scope.

Frontend requirement summary

Progress Invoice

- Contract-level creation
- Gross Remaining subtotal
- Source grouping and availability
- Draft reservation and locks
- Contract Progress and status
- Saved-document previews

Change Order

- Estimate-style editor
- CO number and Estimate link
- Approval lifecycle
- Contract billing status
- Net Contract change
- PDF and Customer Link

Retainage

- Withholding control
- Conditional tracking block
- Available release limit
- Partial and full release
- No empty state when absent
- Customer release document

Standard Invoice keeps its existing independent copy and conversion behavior.

Customer-facing requirement summary

Document	Customer sees	Customer does not see
Progress Invoice	Contract amount, prior billing, current billing, balance, applicable adjustments	Internal pools, reservations, or allocation mechanics
Change Order	CO identity, linked Estimate, revised scope, Contract change, totals	Internal Contract ledger or source-allocation rules
Retainage Release	One Retainage release line and clear amount	Original work repeated as new scope

Web link and PDF present the same core business information. Conditional totals remain conditional.

Technical details

Only implementation constraints that preserve the business model

Stable lineage

Persist source identity so revisions never redistribute historical billing.

Deterministic money

Use fixed decimal calculations and reconcile displayed components to totals.

Document identity

Preserve Estimate, CO, Invoice, and external accounting identifiers.

Existing systems

Reuse current tax, payment, permission, and QBO sync behavior unless these requirements override it.

API design, schemas, polling, webhooks, and service boundaries remain outside this business handoff.

Open production decisions

The core Contract model is complete. Production QBO rollout still needs product and accounting decisions.

Accounting mappings

Confirm Product/Service and account treatment for Retainage, Retainage release, discounts, and Cost Plus.

Tax alignment

Confirm compatible calculation order and rounding across discounts, Cost Plus, Retainage, negative lines, and supported QBO tax modes.

Conflict resolution

Define safe outcomes for Keep QuickBooks version, overwrite, void, relink, and detach without importing QBO edits into Contract ledgers.

Accounting restrictions

Define permissions and behavior for payments, credits, deposits, closed periods, reconciled transactions, and material versus metadata edits.

Safe default: preserve both systems, block the risky QBO write, and require intentional resolution.

Business logic audit

Area	Result	Audit finding
Standard Invoice	PASS	Independent copy and conversion flow
Progress Invoice	PASS	Contract-based from Estimate acceptance
Change Order	PASS	Revises Contract without rewriting history
Retainage	PASS	Withholding and release remain outside scope completion
Payment Schedule	PASS	Planning layer; actual tracked billing remains authoritative
QBO sync	PASS	Accounting destination with conflict safeguards

Core flows are internally consistent. Open QBO production decisions are listed in the preceding appendix.